

Customer Shopping Behavior Analysis

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.



DATASET OVERVIEW

Dataset Summary & Structure

The dataset contains 3,900 rows and 18 columns capturing customer demographics, purchase details, and shopping behavior. Missing data was limited to 37 values in the Review Rating column.

Customer Demographics

Age, Gender, Location, Subscription Status

Purchase Details

Item, Category, Amount, Season, Size, Color

Shopping Behavior

Discount, Promo Code, Previous Purchases, Frequency, Review Rating, Shipping Type

Exploratory Data Analysis in Python

We began with data preparation and cleaning in Python. The dataset was imported using `pandas`, with `df.info()` and `.describe()` used for initial exploration and summary statistics.

Missing values in the Review Rating column were imputed using the median rating of each product category. Columns were renamed to snake case for readability. Feature engineering created `age_group` and `purchase_frequency_days` columns. We verified that `discount_applied` and `promo_code_used` were redundant, dropping the latter. The cleaned `DataFrame` was then loaded into PostgreSQL for SQL analysis.



KEY METRICS

Dataset at a Glance

3,900

Total Purchases

Transactional records analyzed

\$59.76

Avg Purchase Amount

Across all transactions

44

Avg Customer Age

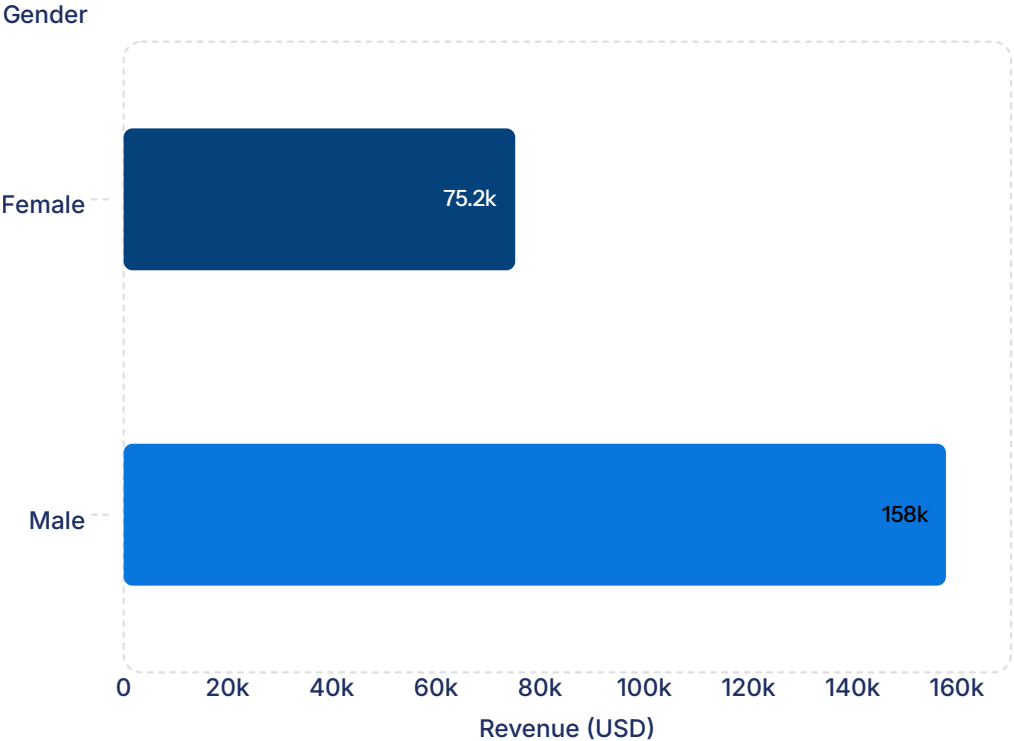
Range: 18–70 years

3.75

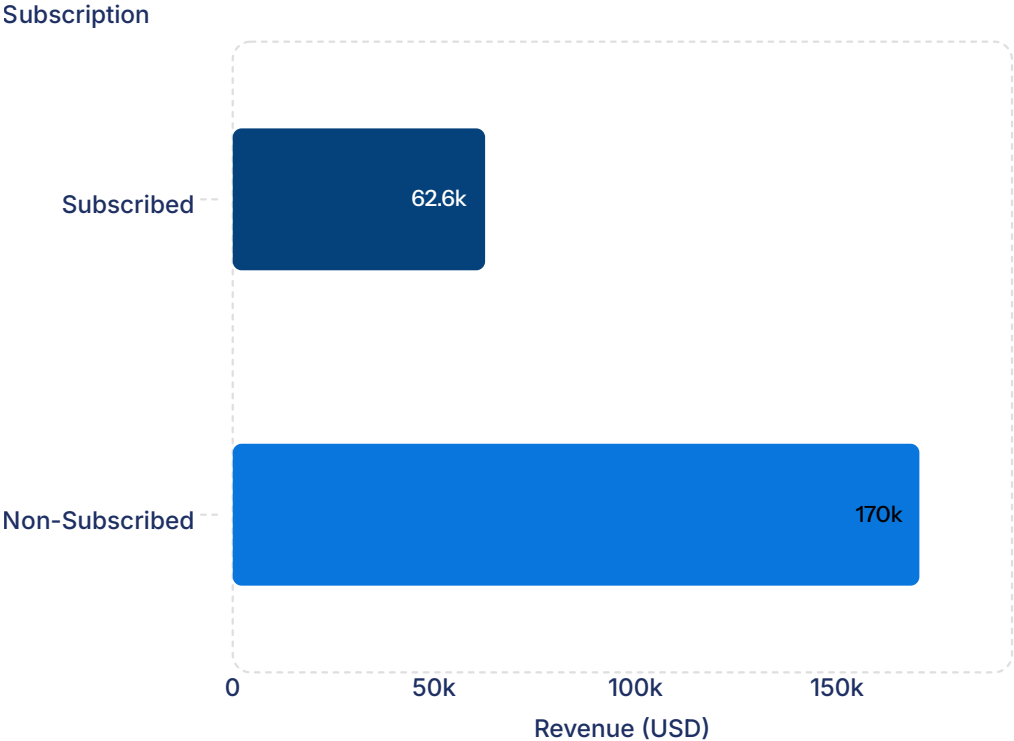
Avg Review Rating

Scale: 2.5–5.0

Revenue by Gender & Subscription Status



Male customers generated \$157,890 in revenue, more than double the \$75,191 from female customers.



Non-subscribers drove \$170,436 in revenue across 2,847 customers, while 1,053 subscribers contributed \$62,645.

Top-Rated Products & Discount-Dependent Items

Top 5 Products by Rating

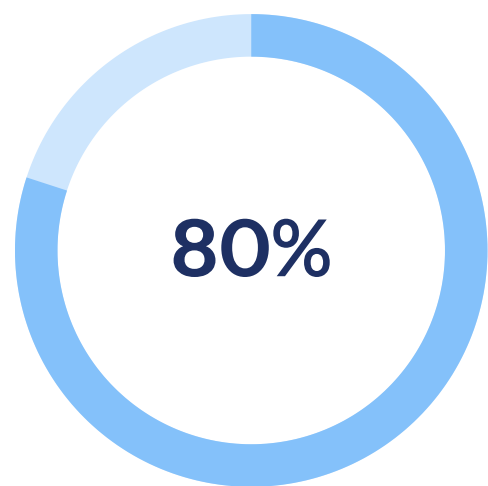
Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Top 5 Discount-Dependent Products

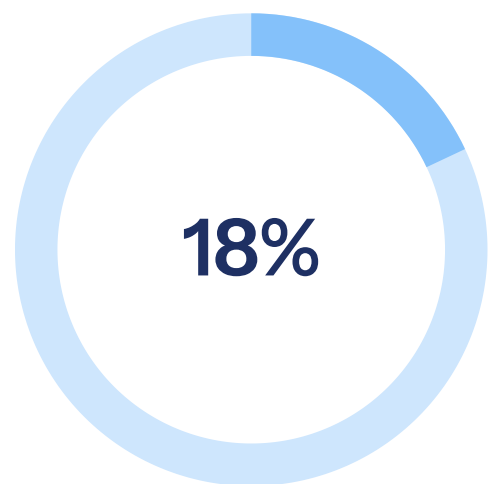
Product	Discount Rate (%)
Hat	50.00
Sneakers	49.00
Coat	49.00
Sweater	48.00
Pants	47.00

Gloves lead in customer satisfaction at 3.86, while Hats are the most discount-dependent at 50% — suggesting promotions heavily drive their sales.

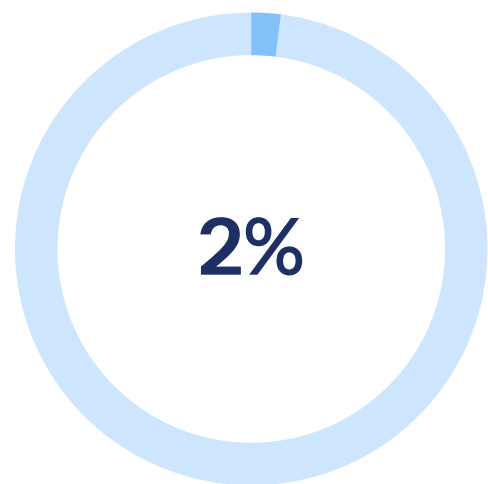
Customer Segmentation & Repeat Buyers



Loyal Customers
3,116 customers



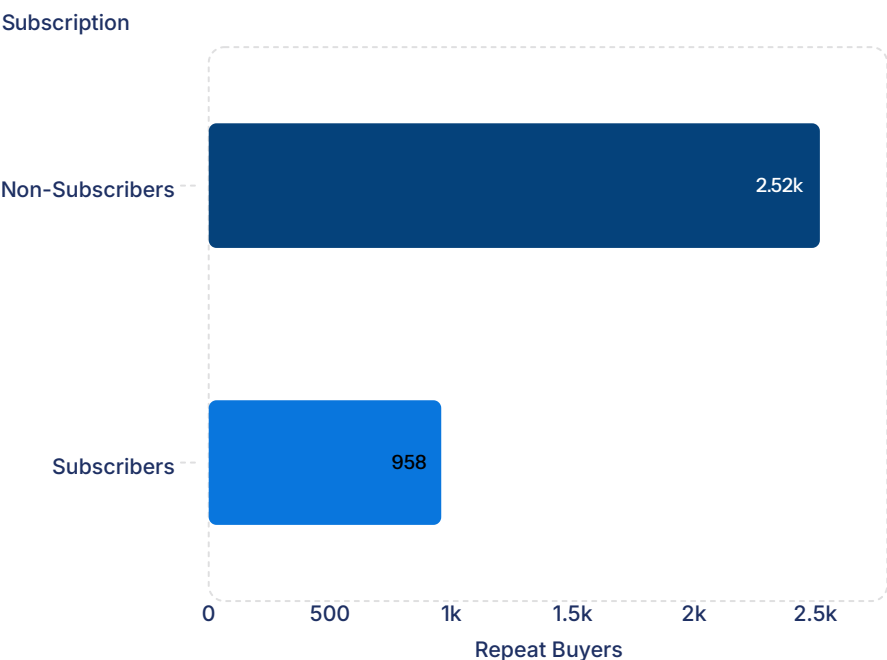
Returning
701 customers



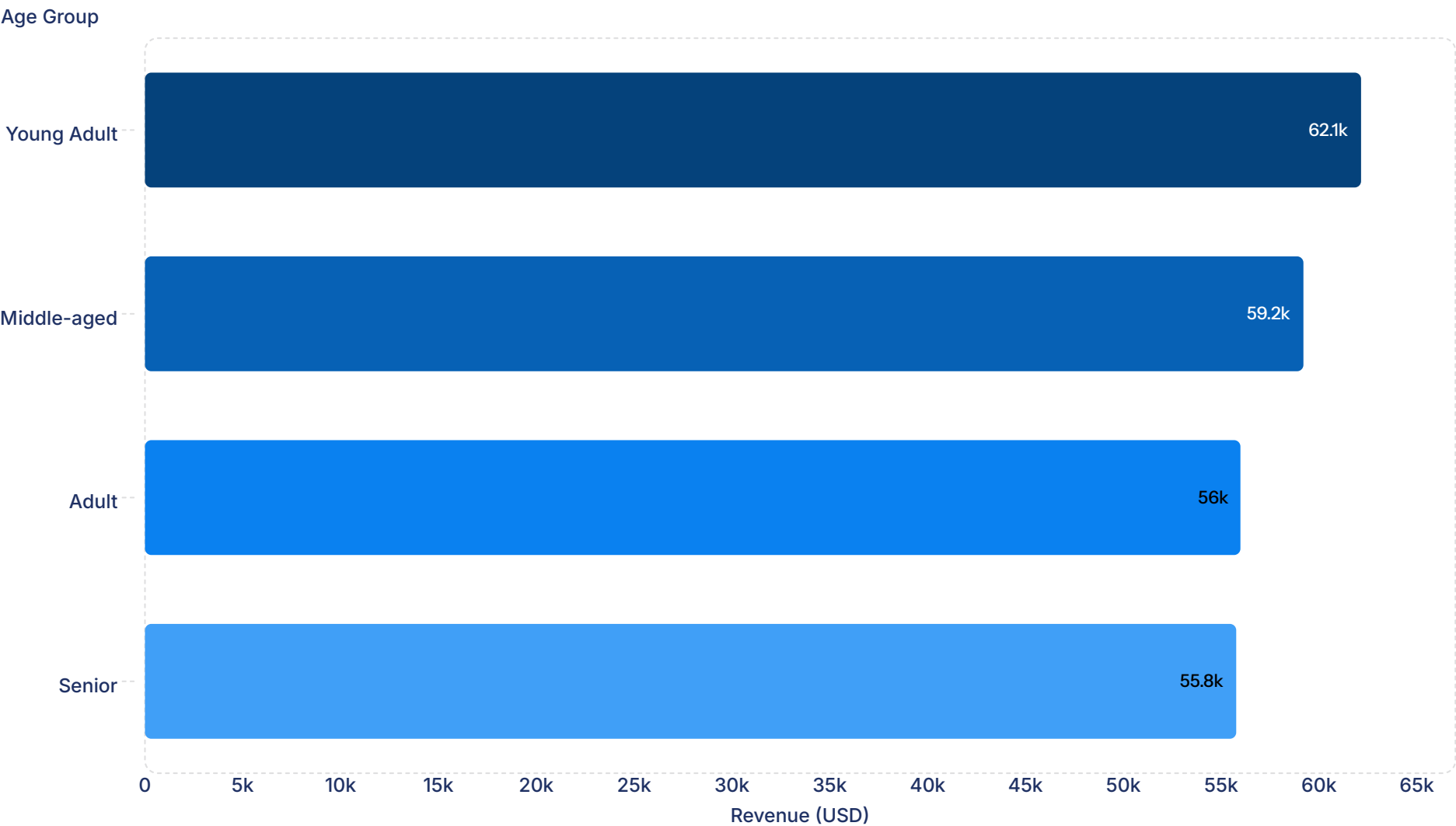
New
83 customers

Repeat Buyers & Subscriptions

Among customers with more than 5 purchases, 2,518 are non-subscribers while 958 are subscribers — indicating repeat purchases do not strongly correlate with subscription uptake.



Revenue by Age Group



Young Adults lead in revenue contribution at \$62,143, followed closely by Middle-aged customers at \$59,197. The spread across age groups is relatively narrow, with Seniors contributing \$55,763.

Power BI Dashboard

We built an interactive dashboard in Power BI to present insights visually. The dashboard features a donut chart for subscriber status (1.05K subscribed vs. 2.85K non-subscribed), revenue and sales breakdowns by category, and revenue and sales by age group.

Key headline metrics displayed include 3.9K total customers, \$59.76 average purchase amount, and a 3.75 average review rating. Interactive filters allow slicing by gender, category, and shipping type.



Business Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers to increase the 27% subscription rate.

Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment and increase retention.

Review Discount Policy

Balance sales boosts with margin control, especially for high-discount products like Hats.

Product Positioning

Highlight top-rated and best-selling products in marketing campaigns.

Targeted Marketing

Focus efforts on high-revenue age groups like Young Adults and express-shipping users.